

We're going to start.'

Day 3 – Part 1: Bren Math Workshop

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Graphs

Why graphs?

Graphs are a way for us to more easily process trends or patterns that may be more challenging to understand in a table or list.

Which could be easier to understand?

The xy coordinate system

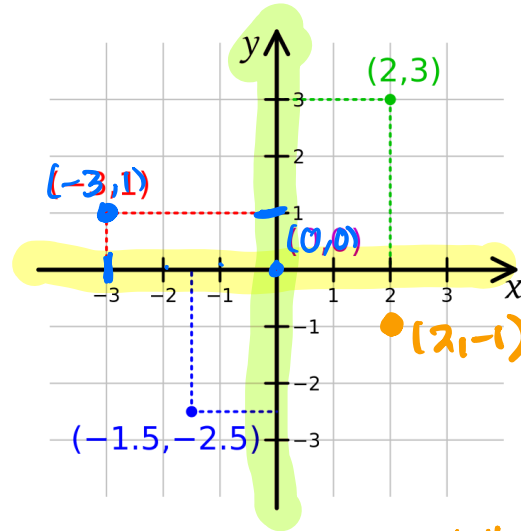
Graphs generally move in a 2-dimensional plane with a coordinate system using two axes:

- x -axis (horizontal axis)
- y -axis (vertical axis)

Axes units must be defined depending on the application.

Use pairs of numbers to place data.

A point on the plane with coords (a,b) means:
the point is located at a on the x axis and
 b on the y axis.



Where would
 $(2,-1)$ be
on the
plane?

The xy coordinate system

* sliders are rendered on website.

We can join series of points to make
graphs

*slides are
viewed on
website

Functions on a single variable and a single output can be graphed

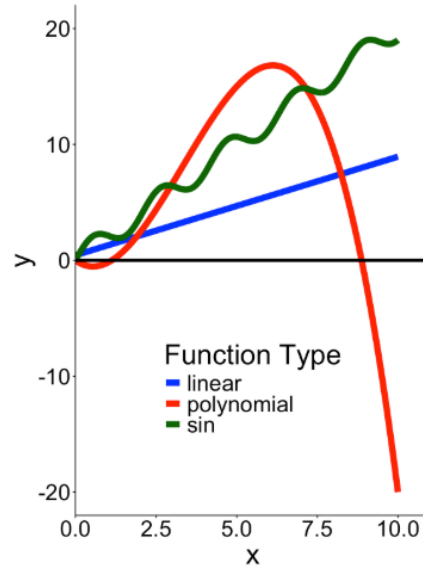
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Two key ingredients to graphs

Intercepts

- x -intercept
 - Where does the graph intersect the x -axis?
 - In other words, for what values of x do we have $(x, 0)$ be part of the graph?

What are the x and y intercepts of the polynomial function in red?

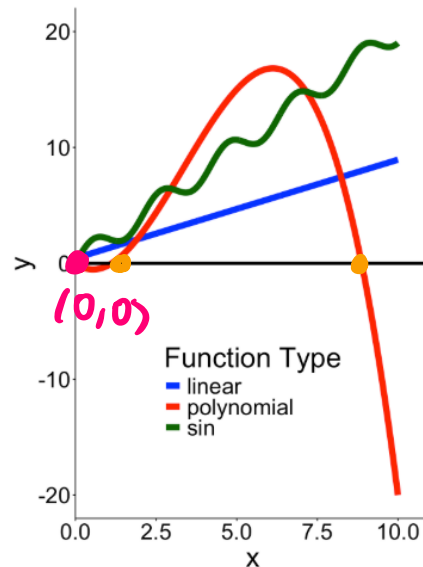


Two key ingredients to graphs

Intercepts

- x -intercept
 - Where does the graph intersect the x -axis?
 - In other words, for what values of x do we have $(x, 0)$ be part of the graph?
- y -intercept
 - Where does the graph intersect the y -axis?
 - In other words, for what value of y do we have $(0, y)$ be part of the graph?

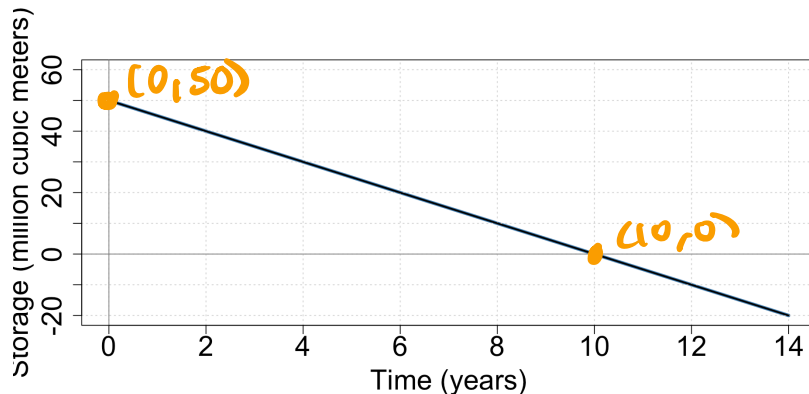
What are the x and y intercepts of the polynomial function in red?



Intercepts practice

 Solve these individually.

1. A reservoir is being drained for irrigation. The graph shows its storage volume (in millions of cubic meters) as a function of time (in years).
- When will the reservoir be empty? Are you using the x or y intercept?
 - What is the initial volume of the reservoir? Are you using the x or y intercept?

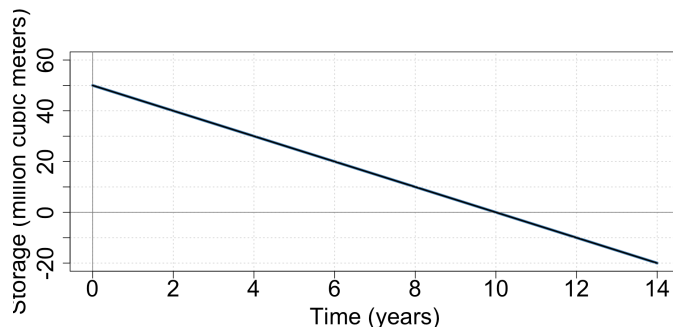


empty in 10 years $\rightarrow x$ intercept
initial @ 50 million cubic meters
 $\rightarrow y$ intercept.

Intercepts practice (1)

- When will the reservoir be empty? Are you using the x or y intercept?
- What is the initial volume of the reservoir? Are you using the x or y intercept?

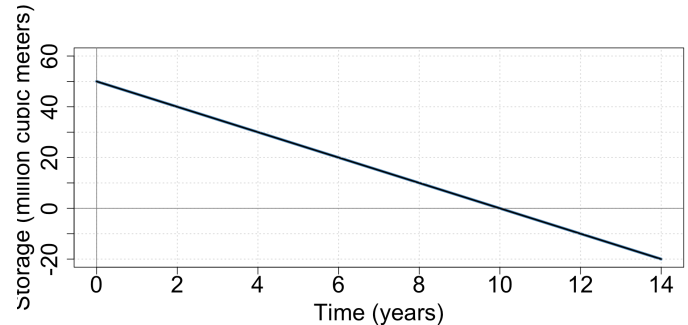
 Let's see a solution!



Intercepts practice (1)

- When will the reservoir be empty? Are you using the x or y intercept?
- What is the initial volume of the reservoir? Are you using the x or y intercept?

 Let's see a solution!



The reservoir will be empty after 10 years. To identify this we look at the x -intercept, which happens at $(10, 0)$.

The initial volume of water in the reservoir is 50 cubic meters. To identify this we look at the y -intercept, which happens at $(0, 50)$.

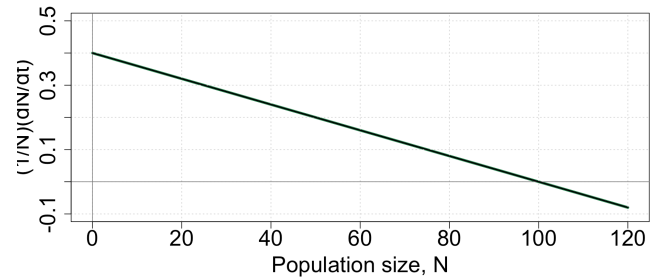
Intercepts practice (2)

Population size is denoted by N , this is the total number of individuals.

Per-capita growth rate is a function of N .

- What does the y -intercept of per-capita growth rate represent?
- What does its x -intercept represent?

 Let's see a solution!



Intercepts practice (2)

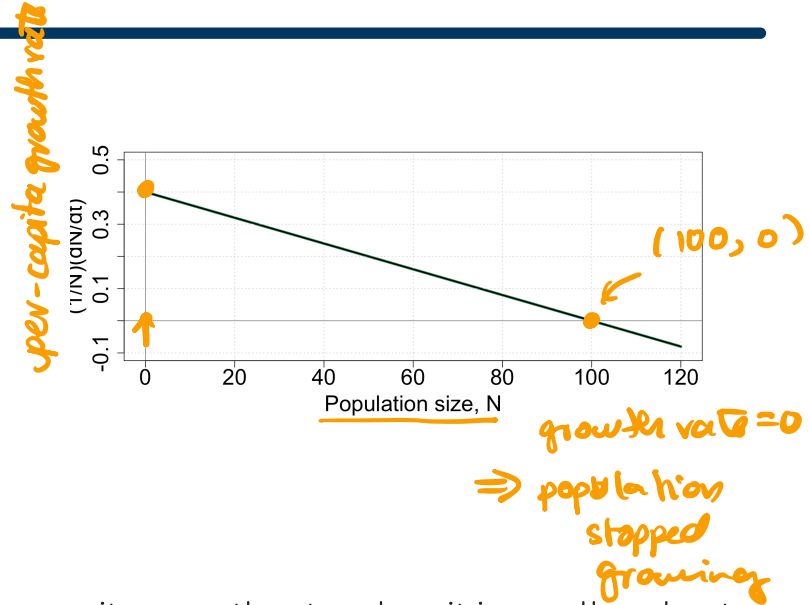
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Per-capita growth rate is a function of N .

- What does the y -intercept of per-capita growth rate represent?
- What does its x -intercept represent?

 Let's see a solution!

- y -intercept = the population's maximum per-capita growth rate when it is small and not yet resource-limited.
- x -intercept = the population size at which per-capita growth stops.



Understanding graphs

When you look at graphs, the first things you should ask:

- What variables are being plotted (e.g. x- & y-axis, including units)?
- What values are plotted (e.g. raw values, transformed, means, etc.)?
- What are the overall takeaways (e.g. trends, maxima, minima, axes intercepts) and am I understanding them responsibly?